BLUESTOCK FINTECH

MUTUAL FUND ANALYTICS CAPSTONE PROJECT

Final Project Report

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Bluestock Fintech for providing me the opportunity to work on the Mutual Fund Analytics Capstone Project. This project enabled me to gain practical experience in data analytics, financial analysis, risk measurement, database management, and dashboard development using industry-standard tools such as Python, SQLite, Jupyter Notebook, and Power BI. I would also like to thank the mentors and team members at Bluestock Fintech for their guidance, support, and valuable feedback throughout the internship period. The knowledge and experience gained during this project have significantly enhanced my analytical, technical, and problem-solving skills.

EXECUTIVE SUMMARY

The Mutual Fund Analytics Capstone Project was undertaken to analyze mutual fund industry trends, evaluate fund performance, assess investment risk, and understand investor behavior using data-driven techniques. The project demonstrates the application of data analytics, financial analysis, and business intelligence tools to derive meaningful insights from large financial datasets.

The project involved the collection, validation, cleaning, and transformation of multiple datasets, including industry Assets Under Management (AUM), Systematic Investment Plan (SIP) inflows, Net Asset Value (NAV) history, benchmark indices, investor transactions, and portfolio holdings. A structured ETL pipeline was developed to ensure data quality and consistency before performing analytical operations.

Exploratory Data Analysis (EDA) was conducted to identify trends in industry growth, SIP adoption, investor demographics, geographic distribution, and sector allocation patterns. Key performance metrics such as Compound Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, Alpha, Beta, and Maximum Drawdown were calculated to evaluate mutual fund performance on both return and risk dimensions.

Advanced analytics techniques were further implemented, including Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio analysis, investor cohort analysis, SIP continuity analysis, sector concentration measurement using the Herfindahl-Hirschman Index (HHI), and a rule-based fund recommendation engine.

To support decision-making and visualization, an interactive Power BI dashboard was developed consisting of four pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. The dashboard enables users to explore key metrics, compare funds, analyze investor patterns, and monitor industry performance through interactive charts and filters.

The outcomes of this project provide valuable insights into mutual fund performance, risk management, portfolio diversification, and investor behavior. The project successfully demonstrates an end-to-end analytics workflow integrating data engineering, statistical analysis, financial modeling, and business intelligence reporting.

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1. INTRODUCTION

1.1. Project Background

The Indian mutual fund industry has experienced significant growth over the last decade due to increasing financial awareness, digital investment platforms, and the widespread adoption of Systematic Investment Plans (SIPs). Mutual funds have emerged as one of the most preferred investment avenues for retail and institutional investors because they provide diversification, professional fund management, and access to various asset classes.

With the rapid growth of the industry, large volumes of data are generated through investor transactions, portfolio holdings, fund performance records, benchmark indices, and industry-wide statistics. Analyzing such data is essential for understanding market trends, evaluating fund performance, assessing risk, and supporting investment decision-making.

Modern data analytics techniques enable organizations and investors to transform raw financial data into actionable insights. This project leverages data engineering, exploratory analysis, performance measurement, risk analytics, and business intelligence tools to develop a comprehensive Mutual Fund Analytics solution.

1.2. Problem Statement

The mutual fund industry generates a large amount of structured financial and transactional data. However, extracting meaningful insights from these datasets remains a challenge due to data quality issues, fragmented information sources, and the complexity of performance evaluation.

Investors often rely solely on returns when selecting funds, while important factors such as risk-adjusted performance, downside risk, portfolio concentration, and investor behavior are overlooked. There is a need for an integrated analytics framework that can combine multiple datasets, evaluate performance using advanced metrics, measure risk, and present findings through interactive visualizations.

1.3. Project Objectives

- Analyze mutual fund industry trends and growth patterns.
- Study Assets Under Management (AUM) and SIP inflow trends.
- Evaluate mutual fund performance using CAGR, Sharpe Ratio, Sortino Ratio, Alpha, and Beta.
- Measure portfolio risk using Maximum Drawdown, VaR, and CVaR.
- Analyze investor demographics and transaction behavior.
- Assess sector concentration risk using HHI.
- Develop a fund recommendation model.
- Build an interactive Power BI dashboard for business users.

1.4. Scope of the Project

The scope of this project includes data ingestion, data validation, cleaning, transformation, exploratory data analysis, performance analytics, advanced risk analytics, investor behavior analysis, dashboard development, and reporting. The analysis is based on historical mutual fund datasets and focuses on generating insights that support investment decision-making and business intelligence reporting.

The project does not attempt to predict future market performance and is limited to the datasets available during the study period.

2. DATA SOURCES

The Mutual Fund Analytics project utilizes multiple datasets covering various aspects of the mutual fund ecosystem, including industry performance, investor transactions, portfolio holdings, benchmark indices, and fund-level performance metrics. These datasets were collected, validated, cleaned, and transformed before being used for analytical and reporting purposes. The integration of multiple datasets enabled a comprehensive analysis of industry trends, fund performance, investor behavior, and portfolio risk.

Table 2.1 Dataset Description

Dataset Name	Description	Purpose
Fund Master Data	Contains scheme information, fund house details, category and expense ratio	Fund Classification
NAV History	Historical daily NAV values for mutual fund schemes	Return and Performance Analysis
Investor Transactions	Investor transaction records including SIPs, redemptions and purchases	Investor Analytics
Portfolio Holdings	Fund-wise stock holdings and sector allocation	Sector Risk Analysis
Industry AUM Data	Assets Under Management across fund houses	Industry Analysis
SIP Inflow Data	Monthly SIP contribution statistics	SIP Trend Analysis
Benchmark Index Data	Nifty benchmark performance data	Alpha and Beta Calculation
Performance Analytics Data	Fund performance metrics and ranking	Scorecard and Ranking Analysis

2.1. Data Characteristics

The datasets consist of both transactional and analytical data. Transactional datasets capture investor activities such as purchases, SIP contributions, and redemptions, while analytical datasets provide information related to fund performance, portfolio composition, and benchmark returns. The data contains both time-series and categorical attributes, enabling trend analysis, risk measurement, segmentation, and comparative evaluation.

The datasets span multiple years and contain information related to mutual fund schemes, investors, sectors, benchmarks, and market performance indicators. Proper preprocessing was performed to ensure consistency and reliability of analytical results.

2.2. Key Data Fields

Field Name	Description
amfi_code	Unique mutual fund Identifier
scheme_name	Name of the mutual fund scheme
fund_house	Asset Management Company
nav	Net Asset Value
daily_return	Daily percentage return
transaction_type	SIP, Purchase, Redemption
amount_inr	Transaction amount
sector	Sector allocation
weight_pct	Portfolio weight percentage
benchmark_return	Benchmark performance metric

2.3. Data Quality Considerations

Prior to analysis, all datasets were subjected to comprehensive validation checks. Missing values, duplicate records, inconsistent formats, and invalid entries were identified and corrected. Date fields were standardized, categorical variables were cleaned, and numerical fields were validated to ensure accuracy. These data quality measures improved the reliability and consistency of analytical outputs generated during the project.

3. PROJECT ARCHITECTURE

3.1. Architecture Overview

The Mutual Fund Analytics Capstone Project follows a structured analytics architecture that transforms raw financial and investor datasets into actionable insights through data processing, performance analytics, risk analysis, and business intelligence reporting.

The architecture consists of multiple stages including data ingestion, validation, transformation, database storage, analytical processing, visualization, and reporting. Each stage plays a critical role in ensuring data quality, analytical accuracy, and effective decision support.

The overall workflow was designed to simulate a real-world financial analytics environment where multiple data sources are integrated and transformed into meaningful business insights.

3.2. System Architecture Diagram

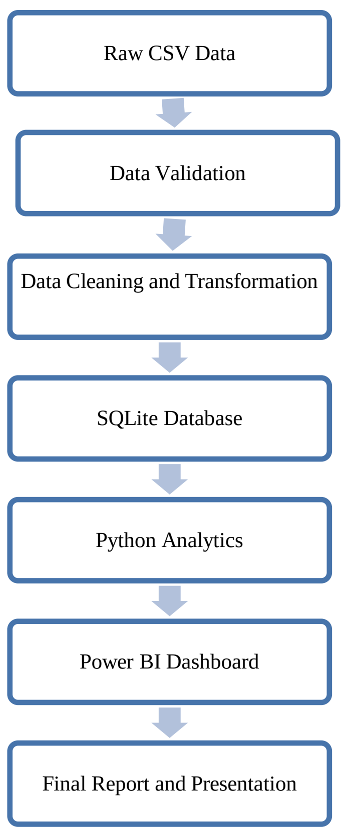

Figure 1: Overall Project Architecture of the Mutual Fund Analytics System

3.3. Technology Stack

Layer	Technology
Programming	Python
Data Processing	Pandas, NumPy
Database	SQLite
Visualization	Matplotlib, Seaborn
Notebook Environment	Jupyter Notebook
Dashboard	Power BI
Version Control	Git
Repository	GitHub

3.4. Project Workflow

The project workflow begins with importing raw datasets from multiple sources. These datasets undergo validation and cleaning to ensure consistency and accuracy. Cleaned data is then stored in SQLite and processed using Python-based analytical workflows.

Exploratory Data Analysis (EDA) is performed to identify trends and patterns across industry metrics, investor behavior, and fund performance. Performance analytics and advanced risk metrics are subsequently calculated to evaluate mutual funds using both return-based and risk-adjusted measures.

The final analytical outputs are visualized through an interactive Power BI dashboard and summarized within a comprehensive project report and presentation.

3.5. Architecture Benefits

- Modular and scalable design
- Supports multiple datasets
- Enables automated analytics workflows
- Facilitates dashboard integration
- Improves reporting efficiency
- Enhances decision-making through data visualization

4. ETL PIPELINE

4.1. Introduction

The ETL (Extract, Transform, Load) pipeline was developed to convert raw mutual fund datasets into a clean and structured analytical dataset. The pipeline ensured data consistency, removed errors, standardized formats, and prepared the data for performance analysis, risk analytics, and dashboard visualization.

The ETL process formed the foundation of the entire project by enabling reliable and efficient data processing workflows.

4.2. Extract Phase

During the extraction phase, multiple datasets related to mutual funds were imported into the analytical environment. These datasets included NAV history, investor transactions, portfolio holdings, SIP inflows, benchmark indices, and fund scorecards.

Python Pandas was used to load the datasets into DataFrames for further processing and validation.

Table 4.1 Source Datasets

Dataset	Purpose
NAV History	Daily return calculation
Investor Transactions	Investor behavior analysis
Portfolio Holdings	Sector concentration analysis
Monthly SIP Inflows	SIP trend analysis
Benchmark Indices	Performance comparison
Fund Scoreboard	Risk and return metrics

4.3. Transform Phase

The transformation phase involved data cleaning, standardization, and feature engineering. Missing values were identified and handled appropriately. Duplicate records were removed, date formats were standardized, and numerical fields were validated for consistency.

Additional analytical features such as daily returns, CAGR metrics, Sharpe Ratio, Sortino Ratio, Alpha, Beta, and risk measures were generated to support advanced analytics.

Table 4.2 Data Cleaning Activities

Activity	Description
Missing Value Handling	Removed or corrected incomplete records
Duplicate Removal	Eliminated duplicate observations
Data Standardization	Unified date formats
Numeric Validation	Checked invalid numerical values
Feature Engineering	Created derived analytical metrics

4.4. Load Phase

After transformation, the cleaned datasets were loaded into an SQLite database. SQLite served as a centralized repository for storing processed datasets and enabled efficient querying during analysis.

The database integration improved data accessibility and supported seamless interaction between Python analytics scripts and Power BI dashboards.

Table 4.2 Data Cleaning Activities

Table Name	Description
nav_history	Historical NAV records
investor_transactions	Investor transaction details
portfolio_holdings	Fund portfolio composition
benchmark_indices	Market benchmark date
monthly_sip_inflows	SIP statistics
fund_scoreboard	Performance metrics

4.5. ETL Workflow Diagram

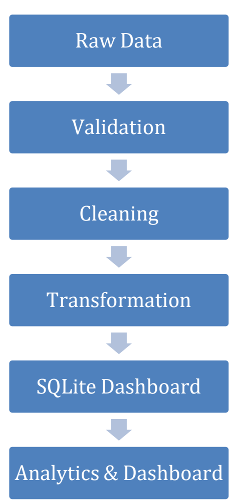

Figure 2: ETL Workflow of the Mutual Fund Analytics Project

4.6. ETL Outcome

- Successfully integrated multiple datasets.
- Improved overall data quality and consistency.
- Generated analytical features for performance evaluation.
- Established a centralized SQLite database.
- Enabled seamless dashboard integration.
- Prepared clean datasets for advanced analytics.

The ETL pipeline ensured that all datasets were accurate, consistent, and analysis-ready. This process significantly improved the reliability of the analytical results and dashboard visualizations developed throughout the project.

5. EXPLORATORY DATA ANALYSIS (EDA)

5.1. Introduction

Exploratory Data Analysis (EDA) was performed to understand industry trends, investor behavior, SIP growth patterns, fund house performance, sector allocation, and portfolio composition. The objective of this phase was to identify meaningful patterns and insights before proceeding to performance and risk analytics.

Various visualizations including line charts, bar charts, and distribution plots were used to analyze the mutual fund ecosystem from multiple perspectives.

5.2. Assets Under Management (AUM) Analysis

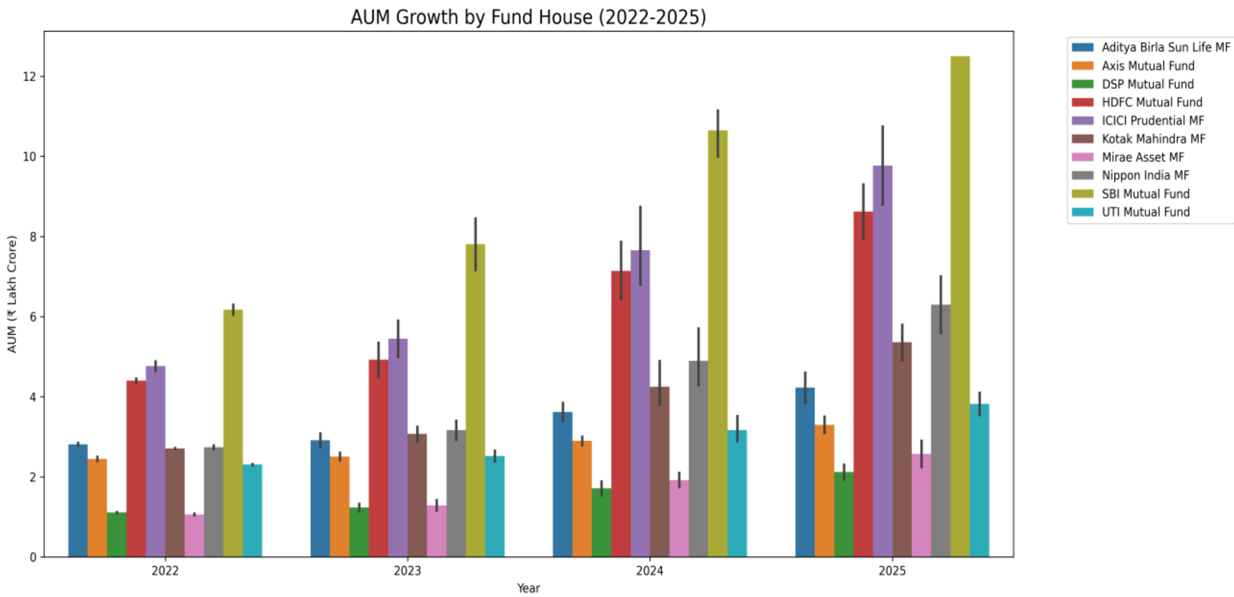

Figure 3: Assets Under Management by Fund House

The AUM analysis revealed significant growth across the mutual fund industry during the study period. Among all fund houses, SBI Mutual Fund maintained the highest Assets Under Management, indicating strong investor confidence and market leadership.

The analysis highlights the increasing adoption of mutual funds as a preferred investment vehicle among retail and institutional investors.

5.3. SIP Trend Analysis

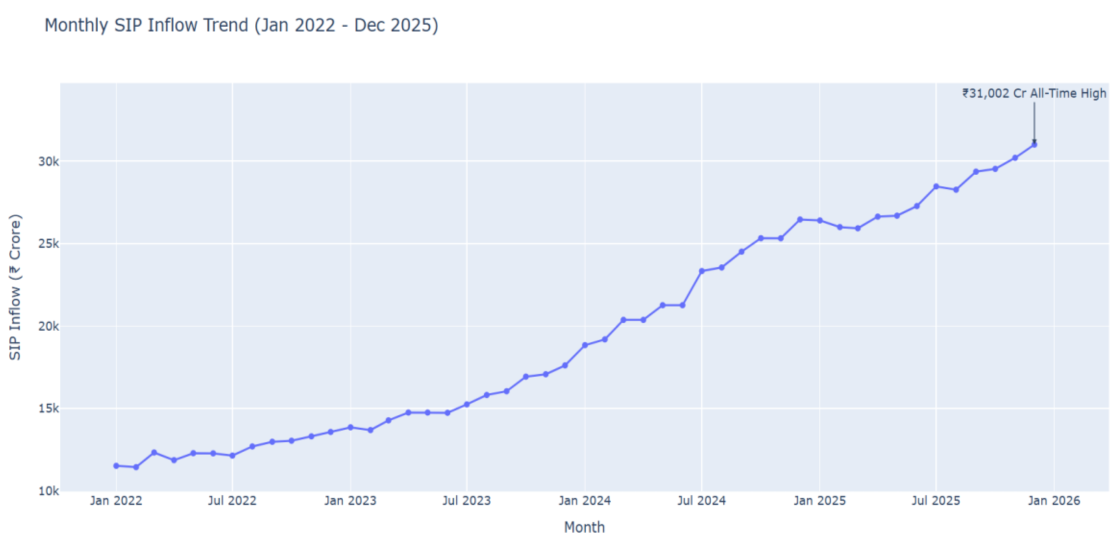

Figure 4: Monthly SIP Inflow Trend (2022-2025)

Systematic Investment Plan (SIP) inflows showed a strong upward trend from 2022 to 2025. The increasing inflows indicate growing investor awareness and long-term investment participation.

The highest monthly SIP inflow was recorded in December 2025, demonstrating sustained confidence in mutual fund investments despite market fluctuations.

5.4. Folio Growth Analysis

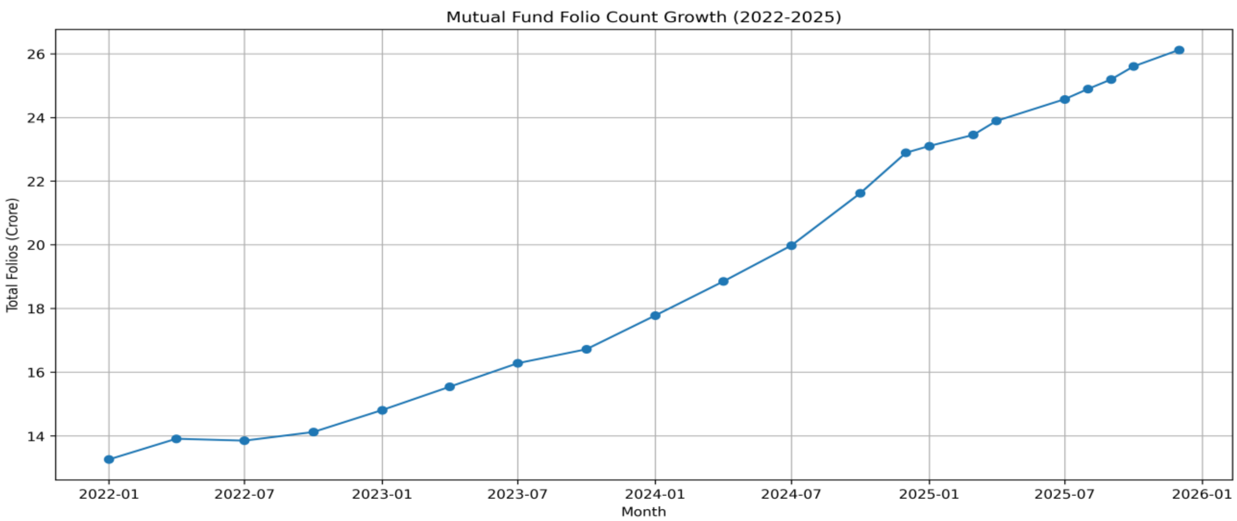

Figure 5: Mutual Fund Folio Growth

Mutual fund folio counts increased consistently throughout the study period. The growth in folios indicates expanding investor participation and broader penetration of mutual fund products across different investor segments.

The near doubling of folio counts during the study period reflects the rapid growth of the Indian mutual fund industry.

5.5. Investor Demographics Analysis

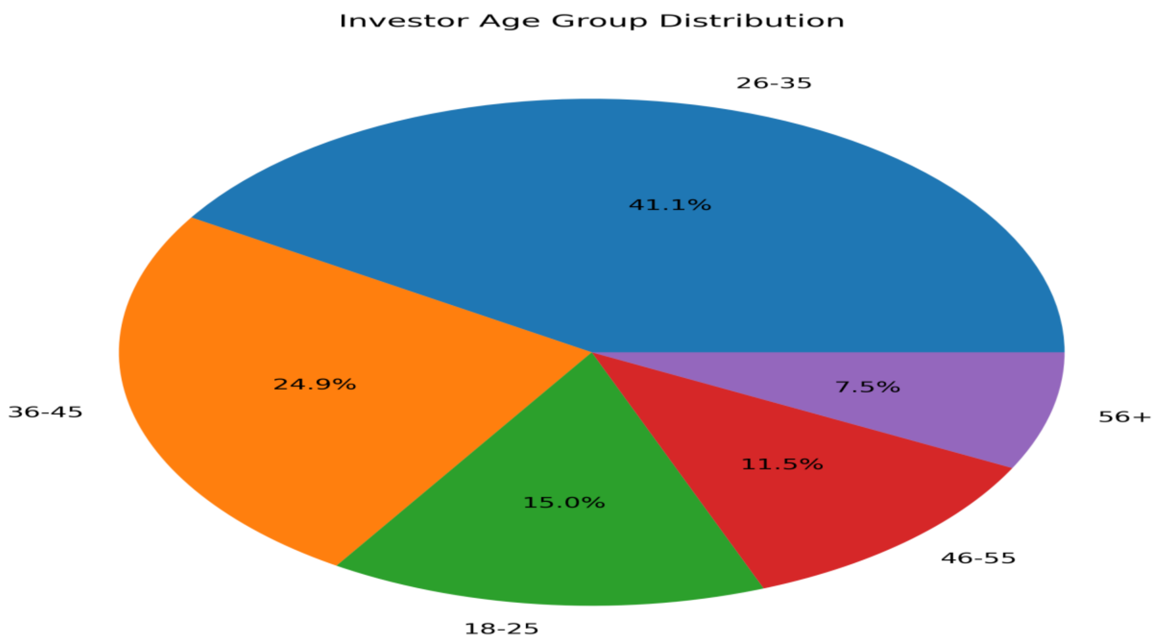

Figure 6: Investor Participation by Age Group

Investor demographic analysis revealed that the 25–34 age group contributed the largest share of mutual fund transactions. This suggests that younger investors are increasingly adopting mutual funds for wealth creation and long-term financial planning.

The analysis also indicates a growing trend of digital investment participation among younger demographics.

5.6. Geographic Analysis

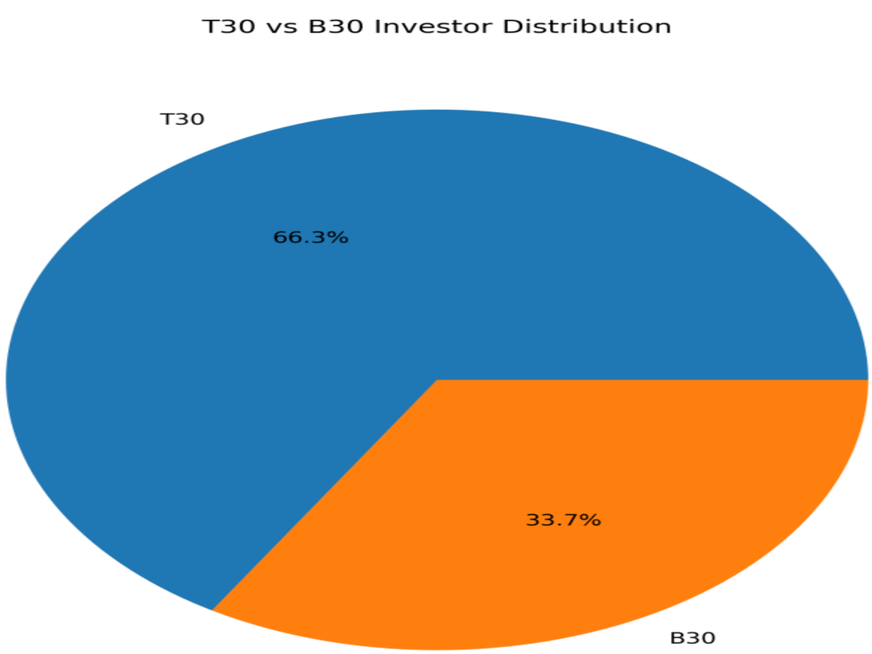

Figure 7: Geographic Distribution of Investors (T30 vs B30)

The geographic analysis indicates that 66.3% of investors belong to T30 cities, while 33.7% belong to B30 regions. This highlights the stronger penetration of mutual fund investments in major urban centers, where financial awareness and investment accessibility are relatively higher. However, the significant contribution from B30 regions demonstrates the growing adoption of mutual funds beyond metropolitan areas, indicating substantial opportunities for future market expansion.

5.7. Sector Allocation Analysis

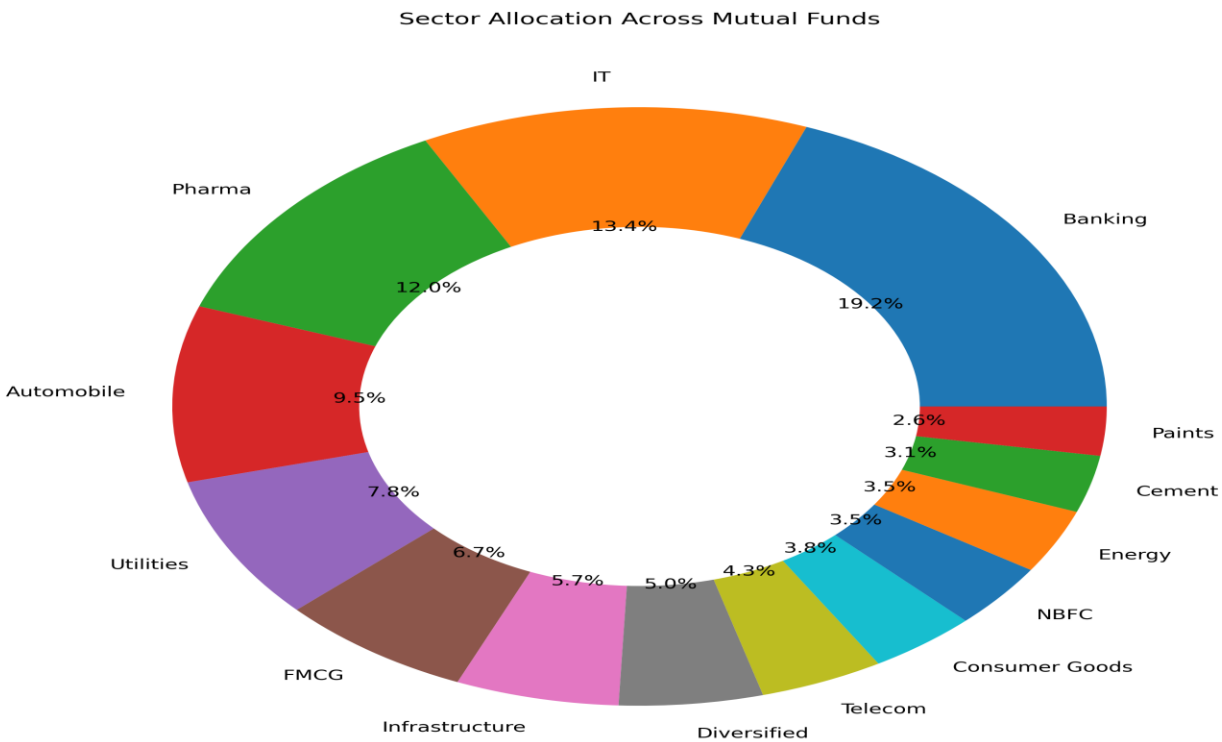

Figure 8: Sector Allocation Across Mutual Fund Portfolios

The sector allocation analysis reveals that the Banking sector accounts for the highest portfolio allocation at 19.2%, followed by Information Technology (13.4%) and Pharmaceuticals (12.0%). These sectors represent a significant portion of mutual fund holdings due to their strong growth potential and market capitalization. The presence of allocations across multiple sectors such as FMCG, Utilities, Infrastructure, Telecom, and Consumer Goods indicates a diversified investment approach. Such diversification helps reduce portfolio risk while maintaining exposure to multiple growth opportunities across the economy.

5.8. Key EDA Findings

- SBI Mutual Fund maintains the highest AUM among all fund houses.
- SIP inflows showed a strong upward trend from 2022 to 2025.

- December 2025 recorded the highest SIP inflow.
- Liquid funds attracted the largest category-wise inflows.
- Mutual fund folio count nearly doubled during the study period.
- SIP remains the dominant investment transaction type.
- Investor participation is highest in the 25–34 age group.
- Tier-1 cities contribute the majority of mutual fund transactions.
- Financial Services and Technology sectors account for a significant share of portfolio allocation.
- Several equity schemes exhibit strong positive return correlations.

6. PERFORMANCE ANALYTICS

6.1. Introduction

Performance Analytics was performed to evaluate mutual fund schemes using various return-based and risk-adjusted performance measures. Key metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Fund Score were analyzed to identify top-performing funds and assess investment quality. These metrics provide a comprehensive view of both returns and associated risks.

6.2. CAGR Analysis

Compound Annual Growth Rate (CAGR) was used to measure the annualized growth of mutual fund investments over different time horizons. CAGR provides a consistent measure of long-term performance and helps compare funds across multiple categories.

amfi_code	cagr_1yr_	cagr_3yr_	cagr_5yr_	scheme_n	fund_hou	category	sub_categ	expense_ratio_pct
100016	-3.30551	-0.03163	2.541248	HDFC Top	HDFC Mut	Equity	Large Cap	1.55
100025	2.50176	4.614329	4.294835	HDFC Sho	HDFC Mut	Debt	Short Dur	0.56
100033	47.73459	33.62921	28.89938	HDFC Mid	HDFC Mut	Equity	Mid Cap	1.38
101206	45.09393	32.52859	22.60479	ABSL Fron	Aditya Birl	Equity	Large Cap	1.6
101207	-24.2421	-2.69648	7.643258	ABSL Smal	Aditya Birl	Equity	Small Cap	1.53

Table 6.1: CAGR Performance of Selected Mutual Funds

The CAGR analysis indicates that equity-oriented schemes delivered significantly higher long-term returns compared to debt funds. Funds such as HDFC Mid-Cap Opportunities Fund and ABSL Frontline Equity Fund demonstrated strong wealth creation potential through superior CAGR values.

6.3. Sharpe Ratio and Sortino Ratio Analysis

Sharpe Ratio and Sortino Ratio were used to evaluate risk-adjusted returns. While Sharpe Ratio considers total volatility, Sortino Ratio focuses specifically on downside risk. Higher values indicate better performance relative to risk.

amfi_code	annual_re	annual_vc	sharpe_ra	sortino_ra	scheme_n	fund_hou	category	sub_categ	expense_ratio_pct
148567	27.03309	14.18762	1.447254	2.382914	Mirae Ass	Mirae Ass	Equity	Large Cap	1.46
120843	27.23652	15.88015	1.305814	2.36162	Kotak Flex	Kotak Mah	Equity	Flexi Cap	1.45
148569	28.30153	17.66639	1.234068	2.144491	Mirae Ass	Mirae Ass	Equity	ELSS	1.6
119551	23.08323	13.73552	1.207324	2.137677	SBI Bluech	SBI Mutua	Equity	Large Cap	1.54
120505	29.23982	19.28263	1.179291	2.027084	ICICI Pru M	ICICI Prud	Equity	Mid Cap	1.36

Table 6.2: Sharpe Ratio and Sortino Ratio Comparison

The analysis revealed that several equity funds maintained higher Sharpe and Sortino Ratios, indicating efficient risk management and superior return generation. These funds provided better compensation for the risks undertaken by investors.

6.4. Alpha and Beta Analysis

Alpha and Beta were analyzed to assess benchmark outperformance and market sensitivity. Alpha measures excess return generated by a fund beyond its benchmark, whereas Beta measures exposure to overall market movements.

amfi_code	alpha_pct	beta	r_squared	scheme_n	fund_hou	category	sub_category
100016	3.100006	0.033602	0.000149	HDFC Top	HDFC Mut	Equity	Large Cap
100025	4.61624	-0.02374	0.001029	HDFC Sho	HDFC Mut	Debt	Short Duration
100033	24.88119	0.167174	0.00217	HDFC Mid	HDFC Mut	Equity	Mid Cap
101206	21.47117	-0.00047	2.86E-08	ABSL Fron	Aditya Birl	Equity	Large Cap
101207	12.97692	-0.16365	0.00112	ABSL Smal	Aditya Birl	Equity	Small Cap

Table 6.3: Alpha and Beta Metrics of Mutual Funds

Positive Alpha values indicate that certain funds outperformed their benchmark indices. Moderate Beta values suggest controlled exposure to market volatility while maintaining competitive returns.

6.5. Maximum Drawdown Analysis

Maximum Drawdown was calculated to measure the largest decline from a portfolio peak to trough. This metric helps investors understand downside risk during adverse market conditions.

amfi_code	max_draw	peak_date	trough_da	scheme_n	fund_hou:	category	sub_category
100016	-24.7344	########	########	HDFC Top	HDFC Mut	Equity	Large Cap
100025	-4.30826	########	########	HDFC Shoı	HDFC Mut	Debt	Short Duration
100033	-16.2172	########	########	HDFC Mid	HDFC Mut	Equity	Mid Cap
101206	-11.2916	########	########	ABSL Fron	Aditya Birl	Equity	Large Cap
101207	-35.4469	########	########	ABSL Smal	Aditya Birl	Equity	Small Cap

Table 6.4: Maximum Drawdown Analysis

Funds with lower drawdown values demonstrated stronger resilience during market corrections. Such funds are generally preferred by risk-conscious investors seeking capital preservation.

6.6. Fund Scorecard Evaluation

A composite fund score was developed by combining multiple performance metrics including CAGR, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown. This score provides a holistic assessment of overall fund quality.

amfi_code	cagr_1yr_	cagr_3yr_	cagr_5yr_	scheme_n	fund_hou:	category	sub_categ	expense_r	sharpe_ra	sortino_ra	annual_vc	alpha_pct	beta	r_squared	max_draw	peak_date	trough_da	rank_3yr_	rank_shar	rank_alph	rank_expe	rank_draw	fund_score
148567	14.5807	31.2786	29.712	Mirae Ass	Mirae Ass	Equity	Large Cap	1.46	1.44725	2.38291	14.1876	26.0441	0.07265	0.00073	-11.2657	#######	#######	0.925	1	0.825	0.45	0.825	84.25
120505	30.3534	30.2065	31.4811	ICICI Pru M	ICICI Prud	Equity	Mid Cap	1.36	1.17929	2.02708	19.2826	30.9529	-0.12109	0.0011	-18.1885	#######	#######	0.9	0.9	1	0.65	0.4	83.25
100033	47.7346	33.6292	28.8994	HDFC Mid	HDFC Mut	Equity	Mid Cap	1.38	1.09292	1.82704	18.9285	24.8812	0.16717	0.00217	-16.2172	#######	#######	0.975	0.85	0.775	0.6	0.525	80.25
120843	28.4811	25.6619	29.6483	Kotak Flex	Kotak Mal	Equity	Flexi Cap	1.45	1.30581	2.36162	15.8802	27.6827	-0.03032	0.0001	-12.974	#######	#######	0.775	0.975	0.925	0.475	0.7	80.25
119094	30.9182	36.066	27.076	Axis Midca	Axis Mutu	Equity	Mid Cap	1.38	0.9975	1.70182	19.3987	28.6505	-0.19931	0.00294	-20.9609	#######	#######	1	0.75	0.95	0.6	0.325	80

Table 6.5: Overall Fund Scorecard Ranking

Funds with the highest composite scores consistently performed well across return, risk, and efficiency metrics. These funds represent balanced investment opportunities for long-term investors.

6.7. Key Performance Findings

- Equity funds generated higher long-term CAGR compared to debt-oriented schemes.
- Top-performing funds exhibited superior Sharpe and Sortino Ratios.
- Positive Alpha values indicate benchmark outperformance.
- Moderate Beta values suggest controlled market sensitivity.
- Lower Maximum Drawdown values indicate stronger downside protection.
- Composite Fund Scores effectively identified high-quality investment options.

7. DASHBOARD OVERVIEW

7.1. Introduction

A Power BI dashboard was developed to visualize industry trends, fund performance, investor behavior, and SIP analytics. The dashboard provides an interactive platform for exploring mutual fund data and deriving actionable insights.

7.2. Dashboard Overview

- ### Dashboard 1: Industry Overview

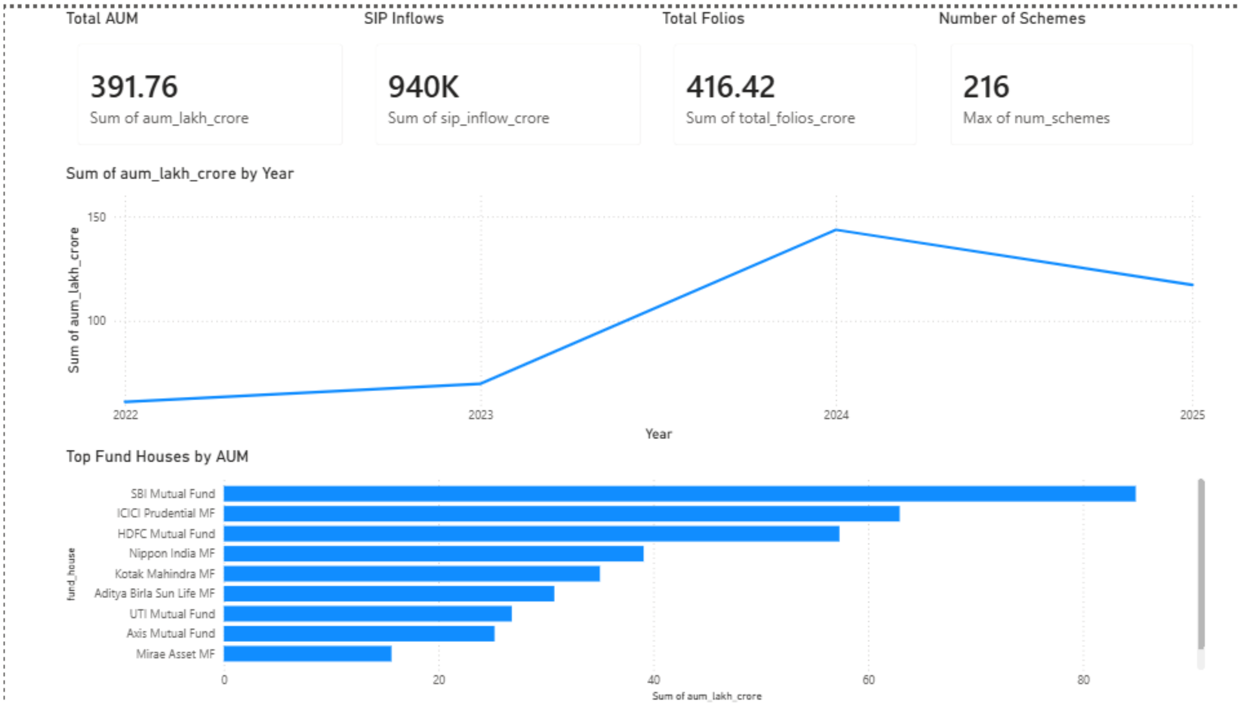

Figure 13: Industry Overview Dashboard

The Industry Overview dashboard presents key industry metrics including Assets Under Management (AUM), SIP inflows, total folios, and the number of mutual fund schemes. It provides a high-level view of overall industry growth and fund house performance.

- ### Dashboard 2: Fund Performance

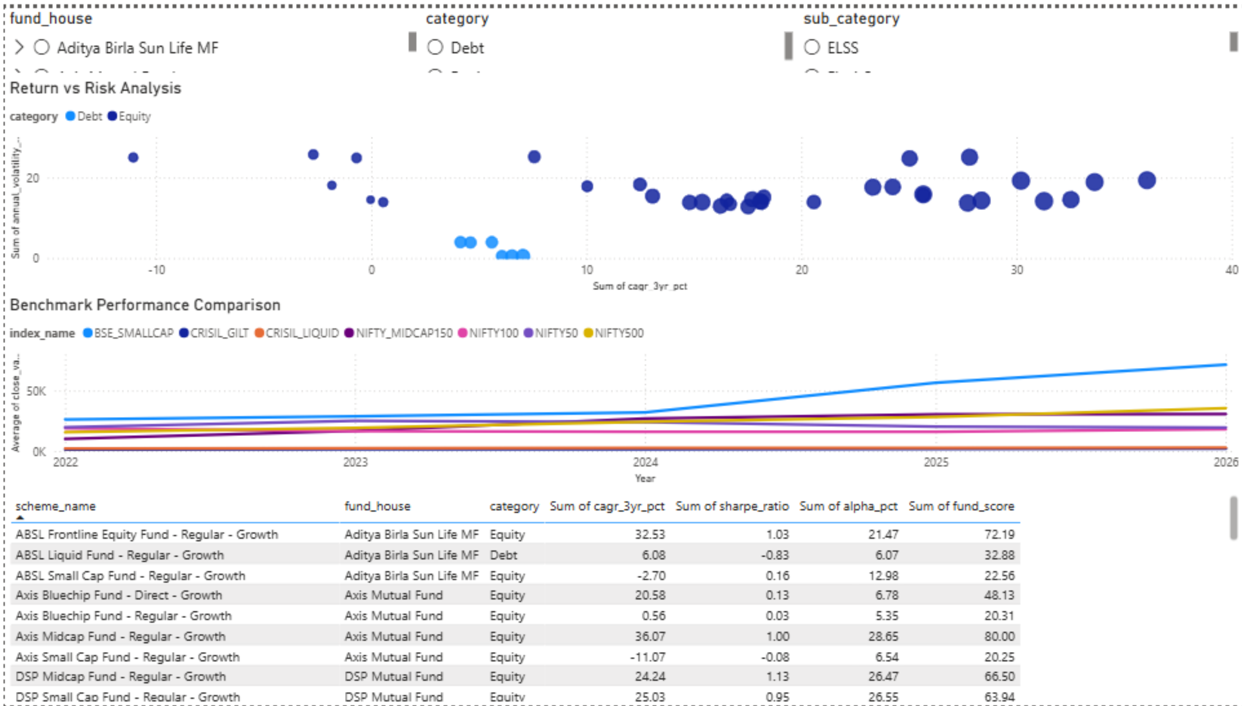

Figure 14: Fund Performance Dashboard

The Fund Performance dashboard evaluates mutual fund schemes using return and risk-based metrics such as CAGR, Sharpe Ratio, Alpha, and Fund Score. Interactive filters allow comparison across fund houses, categories, and sub-categories.

- ## Dashboard 3: Investor Analytics

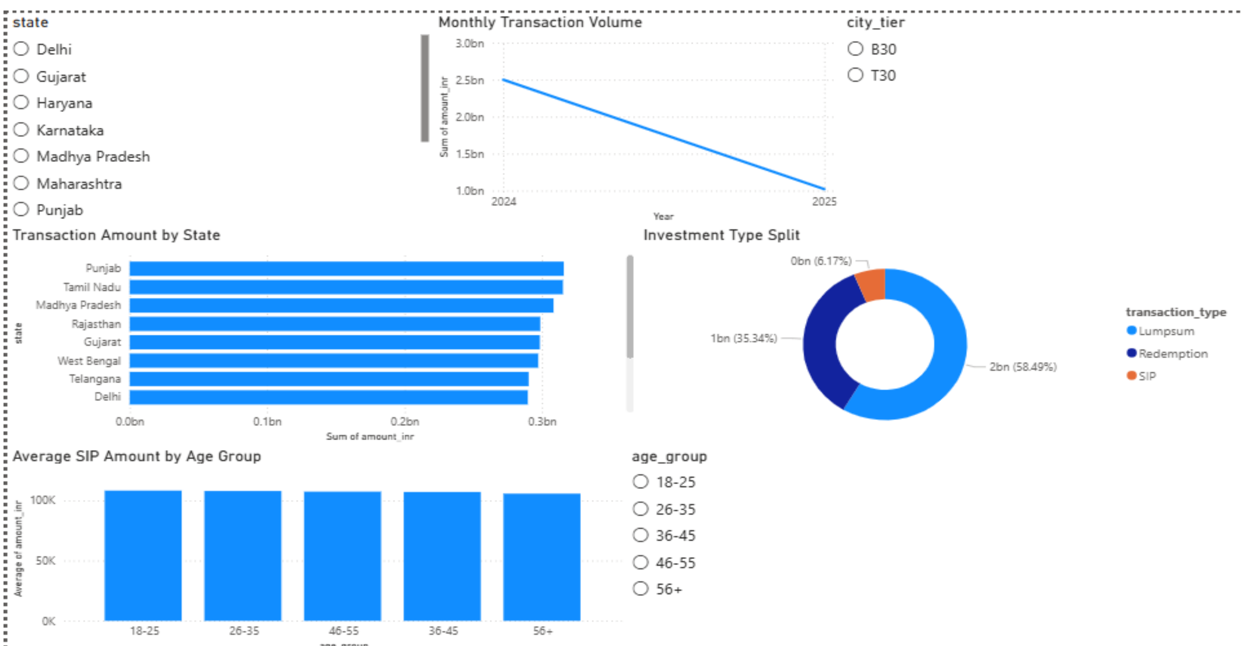

Figure 15: Investor Analytics Dashboard

The Investor Analytics dashboard analyzes investor demographics, transaction patterns, geographic distribution, and investment behavior. It provides insights into age-group participation, transaction volume, and regional investment trends.

8. KEY FINDINGS

- Mutual fund industry exhibited steady growth in AUM and SIP inflows.
- Investors aged 26–35 formed the largest investor segment.
- T30 cities contributed the majority of investment activity.
- Banking, IT, and Pharma sectors dominated portfolio allocations.
- Equity funds delivered superior long-term CAGR performance.
- Risk-adjusted metrics identified several high-quality funds.
- Dashboard analytics enabled efficient monitoring of industry and fund performance.

9. RECOMMENDATIONS

- Investors should evaluate both return and risk metrics before fund selection.
- Diversification across sectors can reduce portfolio concentration risk.
- SIP-based investing remains an effective long-term wealth creation strategy.
- Fund managers should continuously monitor sector concentration and drawdown risk.
- Expanding mutual fund awareness in B30 regions can increase investor participation.

10. LIMITATIONS

- The analysis is based on historical data and may not predict future market performance.
- Some datasets were simulated or limited in scope.
- Market conditions and investor behavior may change over time.
- The recommendation model is rule-based and not driven by machine learning algorithms.

11. CONCLUSION

This project successfully developed a comprehensive Mutual Fund Analytics system using Python, SQLite, and Power BI. The solution integrated data engineering, exploratory analysis, performance evaluation, and dashboard visualization into a unified analytical framework. The findings generated through this project provide valuable insights into investor behavior, fund performance, risk management, and industry trends. Overall, the project demonstrates how data analytics can support informed investment decision-making in the mutual fund industry.

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